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WAFER POSITIONING SYSTEM AND METHOD

Abstract

A semiconductor processing apparatus and a method are provided. The method comprises: placing, by a robot, a wafer on a first supporting area of a lower chamber of a semiconductor processing device according to set or updated current position configuration parameters; etching an edge area of the wafer using the semiconductor processing device; performing image acquisition and recognition to the etched wafer placed based on the current position configuration parameters to obtain etched edge information of the wafer; calculating position deviation parameters using a positioning calibration module; and updating, by a control unit, the current position configuration parameters based on the position deviation parameters, and controlling the robot to place the obtained wafer on the first supporting area of the lower chamber at a position corresponding to the updated current position configuration parameters. This enables improved wafer positioning through calibrating the robot.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of the international application PCT/CN2023/130222, filed on Nov. 7, 2023, which claims priority benefit to Chinese Patent Application No. 202211516242.7, filed on Nov. 29, 2022. The contents of the above-mentioned applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

[0002] The present disclosure relates to the field of wafer positioning, and particularly to a wafer positioning system and a wafer positioning method.

BACKGROUND

[0003] Semiconductor wafers are subjected to a number of processing operations to meet the high standards in the semiconductor manufacturing industries. In advanced applications of semiconductor wafers, it is desirable to have edges of semiconductor wafers that are uniform, smooth, damage-free, and polished, which brings challenges to a semiconductor wafer process resulting in its edge surface evenly and precisely etched.

[0004] FIG. 1a illustrates a top view of a schematic diagram of a semiconductor wafer **100**. Semiconductor wafer **100** comprises a substrate layer **101** and a thin layer **102** deposited on substrate layer **101**. FIG. 1b illustrates a cross-section view of semiconductor wafer **100** along a direction of A-A. Measurement points 1-8 are positions measuring related data of the semiconductor wafer in the processing operations. The etched width refers to the difference between the radius of the substrate layer **101** and the thin layer **102**, as illustrated in FIG. 1b. The etched width shall be substantially uniform at each of the measurement points 1-8. The smaller the difference between the maximum and minimum etched widths, the higher the uniformity. For example, when the edged width is designed for 0.7 mm, the difference between a maximum etched width and a minimum etched width shall not be more than 0.1 mm, otherwise, the etching uniformity will not meet the required standards. The difference between the maximum etched width and the minimum etched width, if exceeding 0.1 mm, will directly affect the quality of subsequent processing operations and eventually cause poor performance of integrated circuit chips, affecting chip manufacturing yield. The semiconductor wafer **100** shown in FIG. 1a is for illustration purposes only. In practice, most semiconductor wafers are not perfectly circular. For instance, a 6-inch wafer has a wafer flat while an 8-inch wafer has a notch.

[0005] The semiconductor wafer wet processing process has the advantages of a simple principle, flexible process and low cost. There are several conventional methods of etching the semiconductor wafer surface edge. For example, a method of polishing the edge area of the semiconductor wafer has been adopted. It rotates the semiconductor wafer and uses physical friction and chemical gas-liquid combination to remove a thin layer from the substrate layer. The polishing method, however, is mainly used in the manufacturing process of semiconductor wafer with less accuracy requirement and is prone to damaging the retained thin layer as well as the substrate layer. The damaged edges may cause edge slip during thermal processing of the wafer and eventually cause

the wafer to be discarded. A method of sucking the semiconductor wafer with a vacuum tip is also commonly adopted. It uses a vacuum tip to suck on the semiconductor wafer where the thin layer is designed to be retained, exposing the rest part of the thin layer outside the vacuum tip, and then immerses the vacuum tip and the semiconductor wafer entirely in a chemical etching solution to etch away the exposed thin layer. However, the vacuum tip method, however, causes the unsmooth removal of the thin layer and uneven etched widths. Similarly, a method of depositing a pre-trimmed thin layer onto the substrate layer of the wafer will cause the uneven etched widths, as the center of the pre-trimmed thin layer may not be aligned with the center of the substrate layer of the wafer. Another commonly used method is the filming method, which uses pure anticorrosive PTFE, PE and other plastic films to protect the parts of the film that need to be retained, and then exposes the whole to a chemically corrosive gas environment or soaks in a chemical corrosive solution to corrode the exposed parts. The filming method is often because the center of the pre-cut film may not be aligned with the center of the wafer substrate, resulting in uneven etching width; and there are many process steps, which need to use a variety of equipment to complete, including filming, wet etching, cleaning and removal of film and other equipment. Similarly, a new method of shower, which works by using a special nozzle to spray the fluid used for etching the edge of the rotating wafer to etching areas, for accurate, uniform, smooth and no damage etching. Although the shower method has a higher etching effect, its design of equipment and parts of the processing precision requirements are very high, the equipment and process cost is too high.

[0006] All advanced wafer edge etching technologies and equipment must address the issue of precise wafer positioning to meet the increasingly stringent requirements for edge etching uniformity and tolerance. This is because during edge etching, the accuracy of the semiconductor wafer positioning directly determines the uniformity of the etched width. Therefore, it is necessary to propose a system and method that can continuously fine-tune and improve the wafer positioning.

SUMMARY

[0007] The implementations of this disclosure provide a wafer positioning system and method capable of continuous improvement, which achieves accurate placement of the wafer at a desired position by repeatedly fine-tuning and calibrating the position configuration parameters of the robot arm.

[0008] To achieve this objective, in one aspect, the present disclosure provides a wafer positioning system. The wafer positioning system comprises: a semiconductor processing device comprising a lower chamber with a first supporting area and an upper chamber with a second supporting area, wherein when the upper chamber engages with the lower chamber with a wafer placed between the first supporting area and the second supporting area, a first channel is formed at an edge area of the first supporting area and/or the second supporting area, and the first channel is configured to provide a first space for flow of one or more chemical fluids for etching an edge area of the wafer; a wafer transfer device comprising a robot and a control unit, wherein the control unit is configured to set or update current position configuration parameters, and the robot is configured to place the fetched wafer on the first supporting area of the lower chamber at a position corresponding to the set or updated current position configuration parameters; an image edge detection device is configured to acquire and recognize images of the etched wafer placed based on the current position configuration parameters to obtain etched edge information of the wafer; a positioning calibration module is configured to calculate position deviation parameters of the current position configuration parameters using the etched edge information obtained through image recognition; wherein the control unit is further configured to update the current position configuration parameters based on the position deviation parameters to obtain the updated current position configuration parameters, and control the robot to place the fetched wafer on the first supporting area of the lower chamber at a position corresponding to the updated current position configuration parameters.

[0009] In another aspect, the present disclosure provides a wafer positioning method. The method

comprises: placing, by a robot, a wafer on a first supporting area of a lower chamber of a semiconductor processing device according to set or updated current position configuration parameters; etching an edge area of the wafer using the semiconductor processing device; performing image acquisition and recognition to the etched wafer placed based on the current position configuration parameters to obtain etched edge information of the wafer; calculating position deviation parameters of the current position configuration parameters using a positioning calibration module with the obtained etched edge information of the wafer; and updating, by a control unit, the current position configuration parameters based on the position deviation parameters to obtain the updated current position configuration parameters, and controlling the robot to place the obtained wafer on the first supporting area of the lower chamber at a position corresponding to the updated current position configuration parameters.

[0010] Compared to the existing technologies, the present disclosure achieves precise wafer positioning by analyzing the etched edge information to iteratively calibrate the positional parameters of the robot. By calculating the position deviation parameters based on the etched edge information and updating the current position configuration parameters accordingly, the system dynamically fine-tunes the placement accuracy of the robot. This ensures that the wafer is placed more precisely on the first supporting area of the lower chamber, resulting in more uniform edge etching.

[0011] It will therefore be understood that this summary is provided solely for purposes of summarizing some exemplary embodiments so as to provide a basic understanding of some aspects of the disclosure. Accordingly, it will be understood that the above described exemplary embodiments are solely illustrative and should not be construed to limiting the scope or spirit of the disclosure in any way. Other example embodiments, aspects, and advantages will become apparent from the following detailed description taken in conjunction with the accompanying figures which illustrate, by way of example, the principles of some described exemplary embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] The present disclosure may be better understood by referring to the drawings as well as the detailed description below. In particular, same numerals are used to refer to same structural parts throughout the drawings, and wherein:

[0013] FIG. 1a illustrates a top view of a schematic diagram of a semiconductor wafer;

[0014] FIG. 1b illustrates a cross-section view of the semiconductor wafer along a direction A-A of FIG. 1a;

[0015] FIG. 2a illustrates a cross-section view of an exemplary device, according to exemplary embodiments of the present disclosure;

[0016] FIG. 2b illustrates a detailed view of a circle A shown in FIG. 2a;

[0017] FIG. 2c illustrates a detailed view of a circle B shown in FIG. 2b;

[0018] FIG. 2d illustrates a detailed view of a circle C shown in FIG. 2c;

[0019] FIG. 2e illustrates a bottom view of an upper chamber of exemplary device;

[0020] FIG. 2f illustrates a top view of a lower chamber of exemplary device;

[0021] FIG. 3 illustrates an exemplary system comprising a semiconductor processing device and a material storage device;

[0022] FIG. 4 illustrates an exemplary method using an device to process an edge area of a semiconductor wafer, according to exemplary embodiments of the present disclosure;

[0023] FIG. 5 illustrates a block diagram of the wafer positioning system in one embodiment;

[0024] FIG. 6 illustrates a 3D schematic of the image edge detection device in one embodiment;

[0025] FIG. 7 illustrates a side view of the image edge detection device in FIG. 6;

[0026] FIG. **8** illustrates a schematic of the edge recognition results from the acquired images;
[0027] FIG. **9** illustrates a comparison between an acquired image of an etched wafer and its HLS image;
[0028] FIG. **10** illustrates a schematic of the multiple lines identified through the edge recognition on an HLS image;
[0029] FIG. **11** illustrates a flowchart of an edge etched detection method for a wafer according to an embodiment of the present disclosure;
[0030] FIG. **12** illustrates a top view of a 6-inch wafer with the edge etched;
[0031] FIG. **13** illustrates a top view of an 8-inch or larger wafer with the edge etched; and
[0032] FIG. **14** illustrates a flowchart of the wafer positioning method in one embodiment.

DETAILED DESCRIPTION

[0033] The following provides a more comprehensive description of certain embodiments of the present disclosure with reference to the accompanying figures, in which some, but not all embodiments of the disclosure are shown. Indeed, various embodiments of the disclosure may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these exemplary embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art. For example, unless otherwise indicated, reference to something as being a first, second or the like should not be construed to imply a particular order. Also, something may be described as being above another item (unless otherwise indicated) may in fact be below it, and vice versa; and similarly, something described as being to the left may be to the right, and vice versa. The same reference numerals refer to the same elements.

[0034] FIGS. **1a** through **1b** illustrate a schematic diagram of a semiconductor wafer **100**. FIG. **1a** illustrates a top view of the schematic diagram of a semiconductor wafer **100**. FIG. **1b** illustrates a cross-section view of semiconductor wafer **100** along the direction A-A of FIG. **1a**. As shown in FIGS. **1a-1b**, semiconductor wafer **100** includes a substrate layer **101** and a thin layer **102** deposited on the first side surface of substrate layer **101**, and substrate layer **101** can be partially covered by thin layer **102**. In another embodiment, substrate layer **101** can be fully covered by thin layer **102**. In another embodiment, both sides of surface of substrate layer **101** can be respectively covered by a thin layer **102**.

[0035] In some embodiments, in processing operations of the semiconductor wafer, thin layer **102** shall be removed from substrate layer **101**. For example, as shown in FIGS. **1a-1b**, a radius of substrate layer **101** is smaller than a radius of thin layer **102**, and an etched width refers to a difference between the two radii. Measurement points 1-8 are positions that measure related data of the semiconductor wafer, as illustrated in FIG. **1a**. The etched width shall be substantially the same at each of the measurement points 1-8. The smaller the difference between the maximum and minimum etched widths, the higher the uniformity will be. For example, when the edged width is designed for 0.7 mm, the difference between the maximum etched width and the minimum etched width shall not be more than 0.1 mm. In some embodiments, thin layers covering both sides of a surface of substrate layer **101** shall be partially or completely removed. The etched width for each side of the surface of substrate layer **101** can be the same or different.

Part I

[0036] FIGS. **2a** through **2f** illustrate an exemplary device **200**, according to exemplary embodiments of the present disclosure. FIG. **2a** illustrates a cross-section view of exemplary device **200**. FIG. **2b** illustrates a detailed view of a circle A shown in FIG. **2a**. FIG. **2c** illustrates a simplified detailed view of a circle B shown in FIG. **2b** (omitting through holes). FIG. **2d** illustrates a detailed view of a circle C shown in FIG. **2c**. FIG. **2e** illustrates a bottom view of an upper chamber **220** of exemplary device **200**. FIG. **2f** illustrates a top view of a lower chamber **210** of exemplary device **200**.

[0037] In one exemplary embodiment, device **200** comprises a lower chamber **210** having a first

supporting area **212**. First supporting area **212** may be configured to support a wafer **100** as described above with reference to FIG. **1** and FIGS. **2a** through **2f**. For example, as shown in FIG. **2a**, first supporting area **212** may have an upper surface facing wafer **100**. Wafer **100** may be placed on the upper surface of first supporting area **212**. In some embodiments, device **200** may comprise an upper chamber **220** having a second supporting area **222**. For example, as shown FIG. **2a**, second supporting area **222** may have a lower surface facing wafer **100**. Upper chamber **220** may be engaged with lower chamber **210** to place wafer **100** between first supporting area **212** and second supporting area **222**. For example, upper chamber **220** may move between two positions relative to lower chamber **210**. In the first position, wafer **100** can be loaded to and/or unloaded from first supporting area **212**. In the second position, upper chamber **220** and lower chamber **210** are engaged with each other so that wafer **100** may be fixed by the upper surface of first supporting area **212** and the lower surface of second supporting area **222** and may be accommodated for processing, as shown in FIG. **2a**. In one embodiment, the upper chamber **220** can be driven to move between two relative positions, while the lower chamber **210** remains stationary. Of course, the lower chamber **210** can also be driven to move between two relative positions while the upper chamber **220** remains stationary. Moreover, both the lower chamber **210** and the upper chamber **220** can be driven simultaneously.

[0038] In some embodiments or any combination of preceding exemplary embodiments of device **200**, as shown in FIGS. **2a** through **2c**, device **200** may comprise a first channel **230** formed at a peripheral area of first supporting area **212** or second supporting area **222**. First channel **230** may be configured to provide a first space **232** for the flow of one or more chemical fluids for etching an edge area of wafer **100**. For example, as shown in FIGS. **2a** through **2c**, first channel **230** may be formed at the peripheral area of second supporting area **222** in upper chamber **220**. First channel **230** may be further formed on a lower surface of upper chamber **220**, and an opening of first channel **230** may face wafer **100**. In one embodiment, first channel **230** may be configured to provide a first space **232**, in which one or more chemical fluids may flow to etch the edge area of wafer **100**. For example, as shown in FIGS. **2a** through **2c**, first space **232** may be formed by an internal surface of first channel **230** and wafer **100**. In one embodiment, first channel **230** may be annular and around the edge area of wafer **100**. The entire edge area of wafer **100** may be accommodated into first space **232**. In another embodiment, first channel **230** may be arranged as an arc with a radian less than 360 degrees, and an edge area of wafer **100** may be selectively accommodated into first space **232**. Then, the one or more chemical fluids may etch part of the edge area of the wafer in accordance with the arc of first channel **230**.

[0039] In some embodiments or any combination of preceding exemplary embodiments of device **200**, as shown in FIGS. **2a** through **2c**, upper chamber **220** may comprise a protrusion part **240** configured to press against an edge of wafer **100**. For example, the protrusion part may directly contact an edge of wafer **100**, and further press against the edge of wafer **100**. In some embodiments, as shown in FIG. **2a**, a center axis X-X of wafer **100** may be perpendicular to an upper surface of wafer **100**. A center axis X'-X' of second supporting area **222** may be perpendicular to a lower surface of second supporting area **222**. A protrusion part **240** may be configured to align the center axis X-X of wafer **100** with the center axis X'-X' of second supporting area **222**. For example, when upper chamber **220** is located in a first position, wafer **100** is loaded onto first supporting area **212**. The center axis X-X of wafer **100** may not be aligned to the center axis X'-X' of a second supporting area **222**. During the course that upper chamber **220** moves from a first position to a second position, protrusion part **240** may contact an edge of wafer **100**, and then may press against an edge of wafer **100**, pushing wafer **100** to move on an upper surface of first supporting area **212**. When upper chamber **220** is located in the second position, the wafer may be fixed on the upper surface of first supporting area **212** and the center axis X-X of wafer **100** may be parallel to the center axis X'-X' of second supporting area **222**. In another embodiment, the center axis X-X of wafer **100** may overlap the center axis X'-X' of second

supporting area **222**.

[0040] In some embodiments or any combination of preceding exemplary embodiments of device **200**, protrusion part **240** may be adjacent to second supporting area **222** and extend toward lower chamber **210**. For example, as shown in FIGS. **2a** and **2b**, protrusion part **240** may be connected to second supporting area **222**. Protrusion part **240** may extend toward lower chamber **210** where upper chamber **220** is located in a second position. In one embodiment, protrusion part **240** may be formed next to first channel **230**, as shown in FIGS. **2a** and **2b**. In some embodiments, as shown in FIG. **2a**, the center axis X-X of wafer **100** may be perpendicular to an upper surface of wafer **100** and the center axis X'-X' of second supporting area **222** may be perpendicular to a lower surface of upper chamber **220**. The upper surface of wafer **100** may be parallel to the lower surface of the second supporting area. In one embodiment, when the upper chamber is located in the second position, the upper surface of wafer **100** may overlap the lower surface of second supporting area **222** and the center axis X-X of wafer **100** may overlap the center axis X'-X' of second supporting area **222**.

[0041] In some embodiments or any combination of preceding exemplary embodiments of the device **200**, protrusion part **240** may include a closed loop arranged around wafer **100**. For example, as shown in FIG. **2a**, protrusion part **240** includes a closed loop. The closed loop may be annular and around the entire edge area of wafer **100**. Protrusion part **240** therefore may uniformly press against the edge area of wafer **100** for overlapping the center axis X-X of wafer **100** with the center axis X'-X' of second supporting area **222**. In some embodiments, the closed loop may include an arc with a radian less than 360 degrees, and the edge area of wafer **100** may be selectively resisted by protrusion part **240** for aligning and/or overlapping the center axis X-X of wafer **100** with the center axis X'-X' of second supporting area **222**. In some embodiment, protrusion part **240** may include an open loop.

[0042] In some embodiments or any combination of preceding exemplary embodiments of device **200**, protrusion part **240** includes an inner corner facing towards the center axis X'-X' of second supporting area **222**. For example, as shown in FIG. **2c**, protrusion part **240** may comprise an inner surface **242** inclining at an angle α to a first reference direction Y-Y. The first reference direction Y-Y may be parallel to a lower surface of second supporting area **222**. The angle α may be within a range of 20°-90°. The inner corner may be formed by inner surface **242** and an inner surface of first channel **230** and may face towards the center axis X'-X' of second supporting area **222**, as shown in FIGS. **2b** and **2c**. In one embodiment, the inner corner may be configured to press against an edge area of wafer **100**. For examples, as shown in FIG. **2b**, during the course that upper chamber **220** moves from a first position to a second position, an inner corner of protrusion part **240** may contact an edge area of wafer **100**, and then may press against an edge of wafer **100**, pushing wafer **100** to move. When upper chamber **220** is located in the second position, the wafer may be fixed and the center axis X-X of wafer **100** may be parallel to the center axis X'-X' of second supporting area **222**. In another embodiment, the center axis X-X of wafer **100** may overlap the center axis X'-X' of second supporting area **222**.

[0043] In some embodiments or any combination of preceding exemplary embodiments of device **200**, a first groove **250** may be formed in a peripheral area **214** of lower chamber **210** and configured to provide a first groove space **252** for the flow of one or more chemical fluids. For example, as shown in FIGS. **2a**, **2b** and **2f**, first groove **250** may be formed in peripheral area **214** of lower chamber **210**, and positioned close to first supporting area **212** of lower chamber **210**. First groove **250** may provide a first groove space **252** and the one or more materials can flow from first space **232** of first channel **230** to first groove space **252**.

[0044] In some embodiments or any combination of preceding exemplary embodiments of device **200**, a passage **260** may be formed between upper chamber **220** and lower chamber **210**. For example, as shown in FIGS. **2b** and **2f**, lower chamber **210** may comprise a first upper surface **262** between first supporting area **212** and first groove **250**. Passage **260** may be formed between first

upper surface 262 of lower chamber 210 and inner surface 242 of protrusion part 240. Passage 260 may connect first space 232 with first groove space 252 for allowing the one or more chemical fluids to flow from first space 232 to first groove space 252 through passage 260. In one embodiment, passage 260 may be blocked by protrusion part 240 from the flow of the one or more chemical fluids from first space 232 to first groove space 252. In another embodiment, passage 260 may be blocked by first supporting area 210 from the flow of the one or more chemical fluids from first space 232 to first groove space 252.

[0045] In some embodiments or any combination of preceding exemplary embodiments of device 200, a first channel 230 may be formed at a peripheral area of second supporting area 222, as shown in FIGS. 2a through 2c. Upper chamber 220 may further comprise a first through hole 270 configured to allow the one or more chemical fluids to flow between a first space 232 and an outside of device 200. For example, first through hole 270 may pass through upper chamber 220 from an outside of device 200 to communicate with first space 232. In one embodiment, the one or more chemical fluids may flow between first space 232 and an outside of device 200 through first through hole 270. In another embodiment, upper chamber 220 may further comprise two or more first through holes (e.g., a secondary first through hole 272, as shown in FIGS. 2a and 2e) which may be substantially the same as the first through hole 270. In this embodiment, at least one first through hole (e.g., first through hole 270) may be configured to serve as an inlet and the rest first through hole(s) (e.g., the secondary first through hole 272) may be configured to serve as an outlet. First space 232 may connect to an outside of device 200 through first through hole 270 and secondary first through hole 272. In this embodiment, the one or more chemical fluids may flow into first space 232 of first channel 230 from the outside of device 200 through first through hole 270, and flow out of first space 232 to the outside of device 200 through secondary first through hole 272.

[0046] In some embodiments or any combination of preceding exemplary embodiments of device 200, a second channel 280 may be formed at a peripheral area of first supporting area 212 and configured to provide a second space 282 for the flow of one or more chemical fluids for etching an edge area of wafer 100. For example, as shown in FIGS. 2a through 2c, a second channel 280 may be formed at a peripheral area of first supporting area 212 in lower chamber 210. Second channel 280 may be further formed on an upper surface of lower chamber 210, and an opening of second channel 280 may face toward wafer 100, as shown in FIGS. 2a and 2f. In one embodiment, second channel 280 may be configured to provide a second space 282, in which the one or more chemical fluids may flow to etch an edge area of wafer 100. For example, as shown in FIGS. 2a through 2c, a second space may be formed by an internal surface of second channel 280 and wafer 100. In one embodiment, second channel 280 may be annular and around an edge area of wafer 100. The entire edge area of wafer 100 may be accommodated into second space 282. In another embodiment, second channel 280 may be arranged as an arc with a radian less than 360 degrees, and the edge area of wafer 100 may be selectively accommodated into second space 282. Then, the one or more chemical fluids may etch a part of the edge area of the wafer in accordance with the arc of the second channel 280. In some embodiments, second channel 280 may be arranged in a shape identical to first channel 230. In some embodiments, second channel 280 may be arranged close to first upper surface 262 between first supporting area 212 and first groove 250. A passage 260 may be formed between first upper surface 262 of lower chamber 210. Passage 260 may connect second space 282 with first groove space 252 for allowing the one or more chemical fluids to flow from second space 282 to first groove space 252 through passage 260. In one embodiment, passage 260 may be blocked by protrusion part 240 from the flow of the one or more chemical fluids from second space 282 to first groove space 252. In another embodiment, passage 260 may be blocked by first supporting area 210 from the flow of the one or more chemical fluids from second space 282 to first groove space 252.

[0047] In some embodiments or any combination of preceding exemplary embodiments of device

200, lower chamber **210** may comprise a second through hole **290** configured to allow one or more chemical fluids to flow between a second space **282** and an outside of device **200**. For example, as shown in FIGS. **2a** and **2b**, second through hole **290** may pass through lower chamber **210** from an outside of device **200** to communicate with second space **282** of second channel **280**. In one embodiment, the one or more chemical fluids may flow between second space **282** and the outside of device **200** through second through hole **290**. In another embodiment, the one or more chemical fluids may flow from the outside of device **200** to second space **282** of second channel **280** through second through hole **290**, and then flow from second space **282** of second channel **280** to first groove space **252** of first groove **250** through passage **260**. In some embodiments, lower chamber **210** may further comprise one or more second through holes (e.g., a secondary second through hole **292**, as shown in FIG. **2a**) which may be substantially the same as second through hole **290**. In this embodiment, at least one second through hole (e.g., a second through hole **290**) may be configured to serve as an inlet and the rest second through hole(s) (e.g., secondary second through hole(s) **292**) may be configured to serve as an outlet. Second space **282** may connect to an outside of device **200** through second through hole **290** and secondary second through hole **292**. In one embodiment, the one or more materials may flow into second space **282** of second channel **280** from the outside of device **200** through second through hole **290**, and flow out of second space **282** to the outside of device **200** through secondary second through hole **292**. In another embodiment, the one or more chemical fluids may flow into second space **282** of second channel **280** from the outside of device **200** through second through hole **290** and secondary second through hole **292**, and then flow from second space **282** of second channel **280** to first groove space **252** of first groove **250** through passage **260**.

[0048] FIG. **3** illustrates an exemplary system **300** comprising a semiconductor processing device **310** and a material storage device **320**, according to exemplary embodiments of the present disclosure. Processing device **310** may be referred to the device **200** as described above with reference to FIGS. **2a-2f**. Processing device **310** may comprise a lower chamber having a first supporting area configured to support a wafer; an upper chamber having a second supporting area, and the upper chamber being engaged with the lower chamber to place the wafer between the first supporting area and the second supporting area; and a first channel formed at a peripheral area of the first supporting area or the second supporting area, the first channel being configured to provide a first space for the flow of one or more chemical fluids for etching an edge area of the wafer. In some embodiments, the upper chamber comprises a protrusion part being configured to press against an edge of the wafer and to align a center axis of the wafer with a center axis of the second supporting area. Material storage device **520** may be connected to processing device **510**. Material storage device may be configured to store the one or more chemical fluids and transfer the one or more chemical fluids between processing device **510** and material storage device **520**. In some embodiments, the one or more chemical fluids may be selected from H_3PO_4 , HF , HCl , HNO_3 , H_2O_2 , or any combination thereof.

[0049] In some embodiments or any combination of preceding exemplary embodiments of system **300**, the protrusion part may be adjacent to the second supporting area and extend toward the lower chamber. The center axis of the wafer may be perpendicular to an upper surface of the wafer, the center axis of the second supporting area being perpendicular to a lower surface of the upper chamber, and the upper surface of the wafer being parallel to the lower surface of the second supporting area. In some embodiments, the protrusion part may include a closed loop arranged around the wafer, and the protrusion part is configured to uniformly press against an edge area of the wafer for overlapping a center axis of the protrusion part with the center axis of the second supporting area.

[0050] In some embodiments or any combination of preceding exemplary embodiments of system **300**, the protrusion part may be adjacent to the second supporting area and extend toward the lower chamber. The center axis of the wafer may be perpendicular to an upper surface of the wafer, the

center axis of the second supporting area being perpendicular to a lower surface of the upper chamber, and the upper surface of the wafer being parallel to the lower surface of the second supporting area. In some embodiments, the protrusion part may include a plurality of juts being circularly and evenly arranged around the wafer to uniformly press against the edge area of the wafer.

[0051] In some embodiments or any combination of preceding exemplary embodiments of system **300**, a first groove may be formed at a peripheral area of the lower chamber and configured to provide a first groove space for the flow of one or more chemical fluids. In some embodiments, a passage may be formed between the upper chamber and the lower chamber, connecting the first space with the first groove space for allowing the one or more chemical fluids to flow from the first space to the first groove space through the passage. In some embodiments, a second groove may be formed at a peripheral area of the upper chamber and positioned above the first groove. In some embodiments, an elastic component may be placed between the first groove and the second groove for blocking the one or more chemical fluids from flowing from the first space to the first groove space.

[0052] In some embodiments or any combination of preceding exemplary embodiments of system **300**, system **300** may comprise a control device **330**. Control device **330** may communicate and control processing device **310** and material storage device **320**. For example, control device **330** can control a move of the upper chamber between a first position of loading/unloading the wafer and a second position of engaging the upper chamber and the lower chamber to process the wafer, the speed of the flow of the one or more chemical fluids, and the direction of the flow of the one or more chemical fluids. Control device **330** can detect the speed of the flow of the one or more chemical fluids, the direction of the flow of the one or more chemical fluids, the condition of the one or more chemical fluids, and the mal-function of processing device **310**. In some embodiments, control device may comprise a PLC, a controller, a sensor, a storage device (e.g., memory, hard drive, SSD, etc.).

[0053] FIG. **4** illustrates an exemplary method **400** using an device to process an edge area of a semiconductor wafer **100**, according to exemplary embodiments of the present disclosure. The method may utilize the device which may be referred to the system as described above with reference to FIGS. **2a-2f**, and **3**.

[0054] In an exemplary embodiment, as shown in FIG. **4**, at step **402**, an device **200** receives and places a wafer on a first supporting area of a lower chamber of the device. At step **404**, the device engages its upper chamber with its lower chamber to place the wafer between the first supporting area and a second supporting area of the upper chamber. At step **406**, a first channel is formed at a peripheral area of the first supporting area or the second supporting area, where in the first channel provides a first space. At step **408**, the device uses a protrusion part to press against an edge of the wafer and to align a center axis of the wafer with a center axis of the second supporting area. At step **410**, the device injects one or more chemical fluids into the first space for etching an edge area of the wafer.

[0055] In some embodiments or any combination of preceding exemplary embodiments of method **400**, at step **402**, a wafer may be conveyed by a wafer conveying device to device **200**. The wafer may further be placed by the wafer conveying device onto a first supporting area of a lower chamber of device **200**. The first supporting area may have an upper surface facing the wafer. The wafer may be placed on the upper surface of the first supporting area. The wafer conveying device may place the wafer onto the first supporting area in a way that part of a lower surface of the wafer is covered by an upper surface of the first supporting area. In some embodiments, the upper chamber of device **200** may be in a first position, where the wafer can be loaded to and/or unloaded from the first supporting area. For example, the wafer can be conveyed from the wafer conveying device to the upper surface of the first supporting area.

[0056] In some embodiments or any combination of preceding exemplary embodiments of method

400, at step **404**, device **200** may engage an upper chamber with its lower chamber to place a wafer between a first supporting area and a second supporting area of the upper chamber. The upper chamber is in a second position where the lower chamber may be engaged with the upper chamber and the wafer may be fixed between the lower chamber and the upper chamber for allowing a process of an edge area of the wafer. The upper chamber may comprise a second supporting area, which may have a lower surface facing the wafer. The upper chamber may be engaged with the lower chamber to place the wafer between the first supporting area and the second supporting area. For example, the wafer may be fixed between the lower surface of the second supporting area and the upper surface of the first supporting area.

[0057] In some embodiments or any combination of preceding exemplary embodiments of method **400**, at step **406**, a first channel may be formed at a peripheral area of a first supporting area. The first channel may be further formed on a lower surface of the upper chamber, and an opening of the first channel may face the wafer. In some embodiments, the first channel provides a first space for allowing a process of an edge area of a wafer. For example, one or more chemical fluids may flow in the first channel and etch an edge area of wafer. In some embodiments, the first channel may be arranged as a closed loop. In some embodiments, the first channel may be arranged as a circle. device **200** or the wafer conveying device may place the wafer in a way that an entire or a partial edge area of the wafer is accommodated into the first space for processing. In some embodiments, the first channel may be arranged as an arc with a radian less than 360 degrees. device **200** or the wafer conveying device may place the wafer in a way that a partial edge area of the wafer is accommodated into the first space for processing.

[0058] In some embodiments or any combination of preceding exemplary embodiments of method **400**, at step **408**, a protrusion part is formed on an upper chamber or a lower chamber of device **200**. The device may use the protrusion part to press against an edge of a wafer. For example, the protrusion part may contact the edge of the wafer during a course that the upper chamber moves from a first position to a second position. Then the protrusion part may press against the edge of the wafer and push the wafer to move on an upper surface of a first supporting area of the lower chamber. When the upper chamber is engaged with the lower chamber, the wafer may be fixed on the upper surface of the first supporting area, and a center axis X-X of the wafer may be aligned with a center axis X'-X' of a second supporting area. A distance between the center axis X-X of the wafer and the center axis X'-X' of the second supporting area may be within a range of 0 mm-0.1 mm. In some embodiments, the protrusion part may be adjacent to the second supporting area and extend toward the lower chamber. In one embodiment, the protrusion part may be formed next to the first channel.

[0059] In some embodiments, the protrusion part includes an inner corner facing towards the center axis X'-X' of the second supporting area. The inner corner may be formed by an inner surface of the protrusion part and an inner surface of the first channel and may face towards the center axis X'-X' of the second supporting area. In one embodiment, the inner corner may be configured to press against an edge area of the wafer. For example, during the course that the upper chamber moves from a first position to a second position, the inner corner of the protrusion part may contact an edge of the wafer, and then may press against the edge of the wafer, pushing the wafer to move. In some embodiments, the inner surface of the protrusion part may contact an edge of the wafer, and then may press against the edge of the wafer, pushing the wafer to move.

[0060] In some embodiments or any combination of preceding exemplary embodiments of method **400**, at step **410**, device **200** may inject one or more chemical fluids into a first space for etching an edge area of a wafer. The one or more chemical fluids may flow around an edge of the wafer in a first space and etch the edge area of the wafer accommodated into the first space. In some embodiments, the device may comprise a through hole connecting the first space with an outside of the device. The one or more chemical fluids may be injected into the first space through the through hole. In some embodiments, the one or more chemical fluids may flow from the first space

into the outside of the device through the through hole. In some embodiments, the device may comprise two through holes, each of which may respectively connect the first space with the outside of the device. The two through holes may be arranged with a distance away to each other. The one or more chemical fluids may be injected into the first space through one through hole and flow from the first space into the outside of the device through the other through hole.

Part II

[0061] By adopting the chamber center positioning method shown in FIGS. 2a-2f, the chamber mechanism and the protrusion part are used to align the wafer with the center of the chamber. However, in some cases, the protrusion part **240** may clamp the wafer or prevent the portion of the wafer in contact with the protrusion part **240** from being etched. Therefore, it is necessary to propose a new wafer positioning solution to solve this problem. In this new solution, the protrusion part **240** may not be provided for wafer positioning, or the protrusion part **240** may be provided simultaneously for wafer positioning.

[0062] FIG. 5 illustrates a block diagram of the wafer positioning system according to the present disclosure. As FIG. 5 has shown, the wafer positioning system comprises a semiconductor processing device **610**, an image edge detection device **500**, a positioning calibration module **620**, and a wafer transfer device **630**. The semiconductor processing device **610** may be the semiconductor processing device **200** described in Part I for etching the edge of the wafer **100**. The semiconductor processing device **610** may be a modified embodiment of the semiconductor processing device **200**, such as removing the protrusion part in semiconductor device **200**, which was originally used for positioning wafer **100**. In some occasions, the positioning calibration module **620** may be integrated with the image edge detection device **500** or with the wafer transfer device **630**. Additionally, the positioning calibration module **620**, the image edge detection device **500**, and the wafer transfer device **630** may be fully integrated together.

[0063] The image edge detection device **500** is configured to perform rapid image detection on the edge etching effect of the etched wafer **100** for immediate evaluation of the edge etching effect. The wafer transfer device may be used to transfer the etched wafer **100** from the semiconductor processing device **610** to the image edge detection device **500**.

[0064] FIG. 6 illustrates a 3D schematic of the image edge detection device **500** in one embodiment. FIG. 7 illustrates a side view of the image edge detection device **500** in FIG. 6. FIG. 8 illustrates a schematic of the edge recognition results from the acquired images.

[0065] As shown in FIGS. 6 and 7, the image edge detection device **500** comprises: a line-scan camera **510**, an illumination system **520**, a rotating stage **530**, a positioning unit **540**, an alignment unit (not labeled), and an image processing unit (not shown).

[0066] The rotating stage **530** is configured to hold the wafer **100** and drive the wafer **100** to rotate around the rotation center. The alignment unit ensures the concentricity between the center of the wafer and the rotation center of the rotating stage. The line-scan camera **510** aligns the edge of the wafer **100** and captures images through line scanning during rotation, stopping when a predetermined height is reached. The image processing unit, electrically connected to the line-scan camera **510**, performs edge recognition on specific regions of the acquired images to identify a first line representing the edge of the etched thin layer and a second line representing the edge of the wafer. The image processing unit may use various existing methods for edge recognition. As shown in FIG. 8, P2 is the thin layer (unetched region), and P1 the substrate layer (edge-etched region), it is clearly shown that there is a sudden change in gray scale at the etched edge (the boundary between etched and unetched regions), so the first line L1 and the second line L2 can be identified based on the change of gray scale. In the present disclosure, by using the line-scanning method, the imaging data can be converted from the Cartesian coordinate system to the polar coordinate system, thereby simplifying the subsequent image processing algorithm of the image processing unit.

[0067] The illumination system **520** provides illumination for the line-scan camera **510**. The

positioning unit **540** identifies the wafer's notch or wafer flat to position the wafer, enabling the line-scan camera **510** to acquire images at specified positions, such as images at the notch or wafer flat where it rotates 90 degrees, 120 degrees, or 180 degrees, etc. This enables fix-point sampling or random sampling.

[0068] When the rotating stage **530** starts rotating, it outputs a synchronization pulse signal to the line-scan camera **510**, which then performs scan sampling. The predetermined height may be set as $1/N$ of the wafer's circumference, where N can be greater than or equal to 1, such as 20, etc. Adjusting N controls the length of the acquired image. To accelerate the detection speed, a larger N can be set.

[0069] The image processing unit detects the distance between the first line **L1** and the second line **L2** and the roughness of the first line **L1**. The distance represents the etched width, while the roughness of the first line **L1** reflects the smoothness of the etched edge. The root mean square error of the first line **L1** may quantify its roughness.

[0070] In practice, the image processing unit also recognizes a third line **L3**, where both the second line **L2** and the third line **L3** represent the edge of the wafer. Due to the curved edge of the wafer, they are recognized as two lines. Typically, the third line **L3** corresponds to the outermost edge of the wafer. In other embodiments, adjusting the illumination system or other parameters may allow detection of only the second line **L2** or the third line **L3**. For example, the distance between the third line **L3** and the first line **L1** may represent the etched width. In this case, only the third line **L3** and the first line **L1** need to be identified. The lines **L1**, **L2** and **L3** can be straight lines or similar straight lines. For example, after magnification, most of the pixels do not form a strict line on the wafer edge, but rather scattered points. However, it overall seems like a straight line.

[0071] The following provides further embodiment details.

[0072] For 12-inch wafers, after the edge etching process is completed, data on the edge oxide layer and etched edge formation will be collected and analyzed to evaluate whether the stability of the device and the final process effect meet the process requirements. Visual detection of edge etching results is critical in the overall wafer processing, it provides strong references for the correction of the overall device's processing flow, thereby improving the function of the processing device. Once any abnormality is found in the process, processing should be stopped to avoid large amount of wafer loss due to either device abnormalities or other reasons.

[0073] A specialized camera and illumination system capture images of the edge of a processed 12-inch wafer, generating a three-level grayscale image distinguishing the etched layer, substrate layer, and the bevel. Image algorithms fit the edges of each layer to assess the edge contour conditions. After the processing procedure, the width of each layer at the wafer edge should fluctuate within a fixed range, and the fluctuation range should not exceed the maximum tolerance specified by the process.

[0074] Currently, the image edge detection device **500** can theoretically achieve a minimum particle size of $3.5\text{ }\mu\text{m}$. However, considering the distortion of the industrial camera used and the noise during the image acquisition process, the actual achievable minimum accuracy is $\pm 10\text{ }\mu\text{m}$. Generally speaking, the image edge detection device **500** can theoretically detect each wafer in 1 minute or less.

[0075] The line-scan camera acquires clear and large-scale images within a very short period of time. After the image acquisition is completed, for the collected images, a dedicated visual processing library is used to perform certain denoising and enhancement on the images. Then, main contours are extracted, and the clear contours are extracted with key points. Using the extracted key points, the least squares method is used to fit the required boundary lines. Finally, the three lines **L1**, **L2** and **L3** that are fitted will be obtained.

Part III

[0076] However, for wafers of different specifications, whether the substrate surface is polished significantly affects the single-channel grayscale values. For example, some unpolished wafers

may have etched region grayscale values between 120-180, while others may range from 70-100; and polished wafers may have etched region grayscale values between 200-255. Thus, the edge detection scheme using grayscale images has cost-increase problems due to different upper and lower limits of grayscale values among different wafers, which requires manual setting for the detection.

[0077] Therefore, in one embodiment, the present disclosure proposes an improved etched edge detection method **700** for a wafer. The etched edge detection method **700** is used to detect wafers with their edges etched, and the etched edge detection method **700** is executed by the image processing unit of the image edge detection device **500**. FIG. **11** illustrates a flowchart of a edge etched detection method **700** for a wafer according to an embodiment of the present disclosure. As shown in FIG. **11**, the method **700** comprises the following steps:

[0078] Step **710**: Acquire the image of the wafer with its edge etched.

[0079] Step **720**: Convert the acquired image to an HLS image.

[0080] In HLS, H stands for hue, L for luminance and S for saturation.

[0081] Whether it is a polished wafer or an unpolished wafer, the etched regions exhibits specific hue values, and the overall color is slightly greenish. FIG. **9** illustrates a comparison between an acquire image of an etched wafer and its HLS image. On the left is the acquired image (i.e., the original image), and on the right is the HLS image, where the etched region is slightly greenish. In this embodiment, the etched region can be confirmed jointly by hue (H) and lightness (L), which has higher accuracy than using grayscale value only.

[0082] Step **730**: In the HLS image, extract pixel points with a luminance value L lower than or equal to a predetermined luminance threshold and a hue value H lower than or equal to a predetermined hue threshold are extracted. After connectivity processing and noise removal, obtain the bevel region and the two side contour lines of the bevel region are obtained. As shown in FIG. **10**, **L11** is the outer edge contour line of the bevel region, and **L12** is the inner edge contour line of the bevel region. For example, if the range of the luminance value and the hue value is 0-255, the predetermined luminance threshold may be 10, and the predetermined hue threshold may be 8 or 0, etc.

[0083] Step **740**: Roughly divide the HLS image into a bevel region **A1**, an etched region **A2**, and an unetched region **A3** based on the luminance values. As shown in FIG. **10**, select a region of a predetermined size within the etched region as an etched region sample, and obtain the upper and lower limits of the luminance L and the upper and lower limits of the hue H based on the etched region sample. The upper and lower limits include both the upper limit value and the lower limit value. As shown in FIG. **10**, the line **L13** represents the etched region sample.

[0084] Step **750**: Binarize the HLS image according to the upper and lower limits of the luminance L and the upper and lower limits of the hue H. Then, obtain the final contour of the etched region through dilation and connectivity processing. The final contour includes the etched boundary line between the etched region and the unetched region. As shown in FIG. **10**, **L14** is the etched boundary line.

[0085] The etched boundary line enables subsequent calculations, such as etched width, etched roughness etc.

[0086] This embodiment of the present disclosure determines the upper and lower limits of H (hue) and L (luminance) of the etched region through a fully automatic sampling method, and then obtains the correct contour of the etched region. There is no need to manually adjust the grayscale values to an appropriate range. Meanwhile, using two channels to obtain the contour is more accurate than using a single channel.

Part IV

[0087] FIG. **12** illustrates a top view of a 6-inch wafer with the edge etched. As shown in FIG. **12**, the wafer comprises a wafer flat **810**, an outer edge contour line **820** at the outermost edge, and an etched boundary line **830** at the inner side. The etched boundary line **830** may be the etched

boundary line **L14** in FIG. **10**, which is obtained through image recognition as described above, and the outer edge contour line **820** may be the line **L11** in FIG. **10**, which is obtained through image recognition as described above. Of course, FIG. **12** only schematically shows the complete outer edge contour line **820** and etched boundary line **830** of the wafer. In specific implementation, the image edge detection device may only obtain parts of the outer edge contour line **820** and the etched boundary line **830** through image recognition. As can be seen from FIG. **12**, if the wafer is precisely centered, the edge of the wafer should be uniformly etched, that is, the etched widths of the 10 measurement points 1-10 in FIG. **12** should be consistent. Therefore, whether the wafer is accurately positioned can be determined by analyzing the etched widths of each measurement point.

[0088] As described above, the wafer positioning system further comprises a positioning calibration module **620** and a wafer transfer device **630**.

[0089] The wafer transfer device **630** comprises a robot and a control unit, wherein the control unit is configured to set or update current position configuration parameters, and the robot is configured to place the fetched wafer on the first supporting area of the lower chamber in the semiconductor processing device **610** at the position corresponding to the set or updated current position configuration parameters. As described above, the semiconductor processing device **610** can etch an edge area of the wafer. As described above, the image edge detection device **500** can acquire and recognize the image of the etched wafer placed based on the current position configuration parameters to obtain etched edge information of the wafer. Specifically, the etched edge can be defined by the outer edge contour line and the etched boundary line, and it certainly can be defined in other ways as well.

[0090] The positioning calibration module **620** is used to calculate the position deviation parameters of the current position configuration parameters by using the etched edge information of the wafer obtained through image recognition. The control unit of the wafer transfer device **630** is also used to update the current position configuration parameters based on the position deviation parameters to obtain updated current position configuration parameters, and control the robot to place the fetched wafer on the first supporting area of the lower chamber at a position corresponding to the updated current position configuration parameters.

[0091] In one embodiment, after the positioning calibration module **620** calculates the position deviation parameters of the current position configuration parameters, it determines whether the position deviation parameters meet the predetermined position deviation range. If so, it stops updating the current position configuration parameters. If not, it updates the current position configuration parameters based on the position deviation parameter. After obtaining the etched edge information of the wafer again with the updated current position configuration parameters, it recalculates the position deviation parameters until the position deviation parameters corresponding to the latest current position parameters meet the predetermined position deviation range.

[0092] In one embodiment, the positioning calibration module **620** calculates the etched widths at multiple measurement points of the wafer using the etched edge information obtained through image recognition, and calculates the position deviation parameters based on the etched widths at the multiple measurement points. FIG. **12** illustrates a top view of a 6-inch wafer with its edge etched. FIG. **13** illustrates a top view of an 8-inch or larger wafer with its edge etched. FIG. **12** shows 10 measurement points, namely measurement points 1-10, and FIG. **13** shows 8 measurement points, namely measurement points 1-8. As shown in FIG. **12**, the wafer has a wafer flat **810**, and as shown in FIG. **13**, the wafer has a notch **850**.

[0093] The position deviation parameter comprises a first direction distance deviation Δx and a second direction distance deviation Δy , while the current position configuration parameter comprises a first direction position coordinate x_1 and a second direction position coordinate y_1 , with the first direction and the second direction perpendicular to each other. The first direction may be the X-axis direction, and the second direction may be the Y-axis direction. Specifically, the first

direction distance deviation Δx of the position deviation parameter refers to the first direction positioning deviation between the center of the wafer and the center of the first supporting area, and the second direction distance deviation Δy of the position deviation parameter refers to the second direction positioning deviation between the center of the wafer and the center of the first supporting area. Based on the first direction position coordinate x_1 in the current position configuration parameter and the first direction distance deviation Δx in the position deviation parameter, the updated first direction position coordinate in the current position configuration parameter is obtained. Based on the second direction position coordinate y_1 in the current position configuration parameter and the second direction distance deviation Δy in the position deviation parameter, the updated second direction position coordinate of the current position configuration parameter is obtained.

[0094] The multiple measurement points comprise multiple pairs of measurement points. As shown in FIG. 12, measurement points 4 and 8 are a pair, measurement points 5 and 9 are a pair, measurement points 2 and 7 are a pair, and measurement points 6 and 10 are a pair. As shown in FIG. 13, measurement points 1 and 5 are a pair, measurement points 2 and 6 are a pair, measurement points 3 and 7 are a pair, and measurement points 4 and 8 are a pair. The straight line connecting each pair of measurement points passes through the center of the wafer, or it can be said that each pair of measurement points are located at both ends of the diameter of the wafer.

[0095] The first direction distance deviation Δx of the position deviation parameter corresponds to one or more pairs of measurement points, and a straight line connecting each pair of measurement points corresponding to the first direction distance deviation passes through the center of the wafer. In the embodiment shown in FIG. 12, the first direction distance deviation Δx corresponds to three pairs of measurement points, namely 4 and 8, 5 and 9, and 6 and 10. In another embodiment, the first direction distance deviation Δx may also correspond to only one pair of measurement points, such as any one of 4 and 8, 5 and 9, and 6 and 10, and it certainly may correspond to more pairs of measurement points as well. Similarly, in FIG. 13, the first direction distance deviation Δx also corresponds to three pairs of measurement points, which will not be described in detail here. The first direction distance deviation Δx is calculated based on the differences in etched widths between each pair of measurement points in the corresponding one or more pairs of measurement points.

[0096] Specifically, as shown in FIG. 12, for example, $\Delta x = |w_5 - w_9|/2$, where w_5 is the etched width of measurement point 5, and w_9 is the etched width of measurement point 9. For another example, $\Delta x = |w_4 - w_8|/2\sqrt{2}$, where w_4 is the etched width of measurement point 4, and w_8 is the etched width of measurement point 8. For another example, $\Delta x = |w_6 - w_{10}|/2\sqrt{2}$, where w_6 is the etched width of measurement point 6, and w_{10} is the etched width of measurement point 10. Of course, Δx can also be calculated by combining w_5 and w_9 , w_4 and w_8 , and w_6 and w_{10} . In this example, Δx is a positive value, and the updated first direction position coordinate in current configuration parameters becomes $x_{1.sub.d} = x_1 \pm \Delta x$. This means when the first direction distance deviation Δx shifts toward one direction of the X-axis, it is $x_{1.sub.d} = x_1 + \Delta x$; while when the first direction distance deviation Δx shifts toward another direction of the X-axis, it is $x_{1.sub.d} = x_1 - \Delta x$. The second direction distance deviation Δy of the position deviation parameters corresponds to one or more pairs of measurement points, and a straight line connecting each pair of measurement points corresponding to the second direction distance deviation passes through the center of the wafer. In the embodiment shown in FIG. 12, the second direction distance deviation Δy corresponds to three pairs of measurement points, namely 4 and 8, 5 and 9, and 6 and 10. In another embodiment, the second direction distance deviation Δy may also correspond to only one pair of measurement points, such as any one of 4 and 8, 5 and 9, and 6 and 10, and it may certainly correspond to more pairs of measurement points as well. Similarly, in FIG. 13, the second direction distance deviation Δy also corresponds to three pairs of measurement points, which will not be described in detail here. The second direction distance deviation Δy is calculated based on the differences in etched widths between each pair of

measurement points in the corresponding one or more pairs of measurement points. Specifically, as shown in FIG. 12, for example, $\Delta y = |w_2 - w_7|/2$, where w_2 is the etched width of measurement point 2, and w_7 is the etched width of measurement point 7. For another example, $\Delta y = |w_4 - w_8|/2\sqrt{\{ \text{square root over } (2) \}}$, where w_4 is the etched width of measurement point 4, and w_8 is the etched width of measurement point 8. For another example, $\Delta y = |w_6 - w_{10}|/2\sqrt{\{ \text{square root over } (2) \}}$, where w_6 is the etched width of measurement point 6, and w_{10} is the etched width of measurement point 10. Of course, Δy can also be calculated by combining w_2 and w_7 , w_4 and w_8 , and w_6 and w_{10} . In this example, Δy is a positive value, and the updated second direction position coordinate in current configuration parameters becomes $y_{1.\text{sub.d}} = y_1 + \Delta y$. This means when the second direction distance deviation Δy shifts toward one direction of the Y-axis, it is $y_{1.\text{sub.d}} = y_1 + \Delta y$; while when the first direction distance deviation Δy shifts toward another direction of the Y-axis, it is $y_{1.\text{sub.d}} = y_1 - \Delta y$.

[0097] Of course, in another embodiment, it can also be defined that both Δy and Δx are positive or negative values. For example, shifting toward one direction is positive values, and shifting toward another direction is negative values. In this case, $y_{1.\text{sub.d}} = y_1 + \Delta y$, $x_{1.\text{sub.d}} = x_1 + \Delta x$.

[0098] It can be seen that the one or more pairs of measurement points corresponding to the first direction distance deviation Δx overlap with the one or more pairs of measurement points corresponding to the second direction distance deviation Δy . Since measurement point 4 is at 45 degrees, measurement point 8 at 225 degrees, measurement point 6 at 315 degrees, and measurement point 10 at 135 degrees, the differences in etched width between the measurement point pairs 4 and 8 and/or the differences in etched width between the measurement point pairs 6 and 10 can be used to calculate both Δx and Δy .

[0099] In one embodiment, the position deviation parameter comprises a positioning angle deviation $\Delta\theta$, and the current position configuration parameter comprises a positioning angle θ . An updated positioning angle in the current configuration parameter is obtained based on the positioning angle θ in the current position configuration parameter and the positioning angle deviation $\Delta\theta$ in the position deviation parameters. For example, the positioning angle of the updated current configuration parameters $= \theta + \Delta\theta$. The positioning angle deviation of the position deviation parameter refers to a positioning angle deviation between the wafer and the first supporting area. For example, as shown in FIG. 12, for a 6-inch wafer, which has a positioning flat **810**, if the positioning angle of the wafer does not match the positioning angle of the first supporting area, the etched width of the positioning flat **810** will also be inconsistent. The positioning angle deviation of the position deviation parameter corresponds to multiple measurement points. As shown in FIG. 12, the positioning angle deviation corresponds to three measurement points, which are measurement points 1, 2, and 3. In other embodiments, the positioning angle deviation can also be calculated by using only two of the measurement points such as 1 and 3. A fitted straight-line slope of the multiple measurement points is calculated based on the differences in etched widths and distances between the corresponding multiple measurement points. The positioning angle deviation $\Delta\theta$ is derived based on the fitted straight-line slope. For example, the distance between measurement points 1 and 3 is D_1 , and the fitted straight-line slope $K = (w_1 - w_3)/D_1 = \tan \alpha$, where w_1 is the etched width of measurement point 1 and w_3 is the etching width of measurement point 3.

[0100] For wafers of 8 inches and above, the wafer does not have a wafer flat but is provided with a notch **850** (see FIG. 13). When the wafer has a notch **850**, the semiconductor processing device **200** will set a guide pin at a corresponding position of the upper chamber **220** or the lower chamber **210**. The guide pin cooperates with the notch **850** to realize the angle positioning of the wafer. If the guide pin still cannot precisely position the wafer at the correct angle, the etched width of the measurement points can also be used to determine the positioning angle deviation. Specifically, the positioning angle deviation of the position deviation parameter corresponds to one or more pairs of measurement points, and each pair of measurement points corresponding to the angle deviation are

located on both sides of the notch **850** of the wafer, The positioning angle deviation is calculated based on the differences in etched widths between each pair of measurement points in the corresponding one or more pairs of measurement points. The larger the positioning angle deviation is, the larger the differences in etched width between the measurement points on both sides of the notch **850** will be.

[0101] In this disclosure, the calibration of the robot comprises: configuring current position configuration parameters; placing the wafer at a position corresponding to the current position configuration parameters for edge etching; recognizing the image of the etched wafer to obtain the etched edge and etched width of the wafer; calculating the position deviation parameter of the current position configuration parameter based on the etched edge of the wafer; determining whether the position deviation parameters meet the predetermined position deviation range; if the position deviation parameters do not meet the predetermined position deviation range, update or calibrate the current position configuration parameters based on the position deviation parameter. In the present disclosure, multiple calibrations may be required to make the position deviation parameters meet the predetermined position deviation range, thereby stopping the wafer positioning. Since each calibration process is the same, it will not be further repeated here.

[0102] In addition, even if the current position configuration parameters have been calibrated and the wafer can be precisely positioned, after a period of time, for example, due to temperature changes, variations in wafer size caused by different wafer batches, or changes in other factors, the quality of the etched edge of the wafer may no longer meet the requirements. At this point, the current position configuration parameters can be recalibrated, that is, the wafer can be repositioned until the quality of the etched edge of the wafer meets the requirements. The key point is that the position deviation parameters meet the predetermined position deviation range.

[0103] In this way, in the present disclosure, the positioning of the wafer can be realized by continuously calibrating the current position configuration parameters of the robot. This solution can cooperate with the protrusion part of the semiconductor processing device **200** mentioned above to position the wafer **100** together, and it is also certain that it can realize the wafer positioning alone.

[0104] Regarding how the robot specifically places the fetched wafer at a position corresponding to the current position configuration parameters according to the current position configuration parameters, this can be achieved by referencing existing technologies. For example, by teaching, an operator can first guide the robot to fetch a wafer from a wafer cassette and transfer the wafer to a specified position on the first support area of the lower chamber in the semiconductor processing device **200**. At this time, the control unit can obtain the current position configuration parameters corresponding to the specified position. For another example, the current position configuration parameters can also be directly set in the control unit.

[0105] In one aspect, the present disclosure provides a wafer positioning method. FIG. **14** illustrates a flowchart of the wafer positioning method **900** in one embodiment. As shown in FIG. **14**, the wafer positioning method **900** comprises the following steps:

[0106] Step **902**: a robot places a wafer on the first supporting area of a lower chamber in a semiconductor processing device according to set or updated current position configuration parameters.

[0107] Step **904**: the semiconductor processing device etches the edge area of the wafer. Refer to first part of the content for specific etching process.

[0108] Step **906**: acquire and recognize the image of the etched wafer placed based on the current position configuration parameters to obtain etched edge information (e.g. the etched width and etched boundary line) of the wafer. Refer to the second and third part of the content for specific image recognition.

[0109] Step **908**: the positioning calibration module calculates position deviation parameters of the current position configuration parameters by using the obtained etched edge information of the

wafer.

[0110] Step **910**: a control unit updates the current position configuration parameters based on the position deviation parameters to obtain the updated current position configuration parameters, and controls the robot to place the fetched wafer on the first supporting area of the lower chamber at a position corresponding to the updated current position configuration parameters.

[0111] In one embodiment, wherein after the positioning calibration module calculates the position deviation parameters of the current position configuration parameters, the method further comprises: determining whether the position deviation parameters meet the predetermined position deviation range. If so, it stops updating the current position configuration parameters. If not, it updates the current position configuration parameters based on the position deviation parameter. After obtaining the etched edge of the wafer again with the updated current position configuration parameters, it recalculates until the position deviation parameters corresponding to the latest current position parameters meet the predetermined position deviation range.

[0112] The other contents of the wafer positioning method **900** can be referred to the corresponding parts in the wafer positioning system described above. Therefore, they will not be further repeated here.

[0113] The above description is intended to be illustrative, and not restrictive. Although the present disclosure has been described with references to specific illustrative examples, it will be recognized that the present disclosure is not limited to the examples described. The scope of the disclosure should be determined with reference to the following claims, along with the full scope of equivalents to which the claims are entitled.

[0114] As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises”, “comprising”, “includes” and/or “including”, when used herein, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. Also, the terms “first,” “second,” “third,” “fourth,” etc., as used herein are meant as labels to distinguish among different elements and may not necessarily have an ordinal meaning according to their numerical designation. Therefore, the terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting.

[0115] It should also be noted that in some alternative embodiments, the functions/acts noted may occur out of the order noted in the figures. For example, two figures shown in succession may in fact be executed substantially concurrently or may sometimes be executed in the reverse order, depending upon the functionality/acts involved.

[0116] Although the method operations were described in a specific order, it should be understood that other operations may be performed in between described operations. Described operations may be adjusted so that they occur at slightly different times or the described operations may be distributed in a system. The system allows the occurrence of the processing operations at various intervals associated with the processing.

[0117] Many modifications and other embodiments of the disclosure set forth herein will come to mind to one skilled in the art to which the disclosure pertains having the benefit of the teachings presented in the foregoing description and the associated figures. Therefore, it is to be understood that the disclosure is not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Moreover, although the foregoing description and the associated figures describe example embodiments in the context of certain example combinations of elements and/or functions, it should be appreciated that different combinations of elements and/or functions may be provided by alternative embodiments without departing from the scope of the appended claims. In this regard, for example, different combinations of elements and/or functions than those explicitly described

above are also contemplated as may be set forth in some of the appended claims. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. A wafer positioning system, comprising: a semiconductor processing device comprising a lower chamber with a first supporting area and an upper chamber with a second supporting area, wherein in response to the upper chamber engages with the lower chamber with a wafer placed between the first supporting area and the second supporting area, a first channel is formed at an edge area of the first supporting area and/or the second supporting area, and the first channel is configured to provide a first space for flow of one or more chemical fluids for etching an edge area of the wafer; a wafer transfer device comprising a robot and a control unit, wherein the control unit is configured to set or update current position configuration parameters, and the robot is configured to place the fetched wafer on the first supporting area of the lower chamber at a position corresponding to the set or updated current position configuration parameters; an image edge detection device is configured to acquire and recognize images of the etched wafer placed based on the current position configuration parameters to obtain etched edge information of the wafer; and a positioning calibration module is configured to calculate position deviation parameters of the current position configuration parameters using the etched edge information obtained through image recognition; wherein the control unit is further configured to update the current position configuration parameters based on the position deviation parameters to obtain the updated current position configuration parameters, and control the robot to place the fetched wafer on the first supporting area of the lower chamber at a position corresponding to the updated current position configuration parameters.
2. The wafer positioning system of claim 1, wherein after the positioning calibration module calculates the position deviation parameters of the current position configuration parameters, the positioning calibration module determines whether the position deviation parameters meet a predetermined position deviation range, if so, the positioning calibration module stops updating the current position configuration parameters, if not, the positioning calibration module updates the current position configuration parameters based on the position deviation parameters, the positioning calibration module recalculates the position deviation parameters after obtaining the etched edge information of the wafer again with the updated current position configuration parameters until the position deviation parameters corresponding to the latest current position parameters meet the predetermined position deviation range.
3. The wafer positioning system of claim 1, wherein the positioning calibration module calculates etched widths at multiple measurement points of the wafer using the etched edge information obtained through image recognition, and calculates the position deviation parameters based on the etched widths at multiple measurement points.
4. The wafer positioning system of claim 3, wherein the position deviation parameters comprise a first direction distance deviation and a second direction distance deviation, while the current position configuration parameters comprise a first direction position coordinate and a second direction position coordinate, with the first direction and the second direction perpendicular to each other; the first direction position coordinate in the updated current position configuration parameters is obtained based on the first direction position coordinate in the current position configuration parameters and the first direction distance deviation in the position deviation parameters, and the second direction position coordinate in the updated current position configuration parameters is obtained based on the second direction position coordinate in the current position configuration parameters and the second direction distance deviation in the position deviation parameters.

5. The wafer positioning system of claim 4, wherein the first direction distance deviation of the position deviation parameter refers to a first direction positioning deviation between a center of the wafer and a center of the first supporting area; and the second direction distance deviation of the position deviation parameter refers to a second direction positioning deviation between the center of the wafer and the center of the first supporting area.
6. The wafer positioning system of claim 4, wherein the multiple measurement points comprise multiple pairs of measurement points, the first direction distance deviation of the position deviation parameters corresponds to one or more pairs of measurement points, and a straight line connecting each pair of measurement points corresponding to the first direction distance deviation passes through the center of the wafer, the first direction distance deviation is calculated based on differences in the etched widths between each pair of measurement points in the corresponding one or more pairs of measurement points, alternatively, the second direction distance deviation of the position deviation parameters corresponds to one or more pairs of measurement points, and a straight line connecting each pair of measurement points corresponding to the second direction distance deviation passes through the center of the wafer, the second direction distance deviation is calculated based on differences in the etched widths between each pair of measurement points in the corresponding one or more pairs of measurement points.
7. The wafer positioning system of claim 6, wherein the one or more pairs of measurement points corresponding to the first direction distance deviation overlap with the one or more pairs of measurement points corresponding to the second direction distance deviation.
8. The wafer positioning system of claim 4, wherein the position deviation parameters further comprise a positioning angle deviation, and the current position configuration parameter further comprises a positioning angle, an updated positioning angle in the current configuration parameter is obtained based on the positioning angle in the current position configuration parameter and the positioning angle deviation in the position deviation parameter, the positioning angle deviation of the position deviation parameter refers to a positioning angle deviation between the wafer and the first supporting area.
9. The wafer positioning system of claim 8, wherein the positioning angle deviation of the position deviation parameters corresponds to multiple measurement points, and a fitted straight-line slope of the multiple measurement points is calculated based on the differences in etched widths and distances between the corresponding multiple measurement points, the positioning angle deviation is derived based on the fitted straight-line slope; or the positioning angle deviation of the position deviation parameters corresponds to one or more pairs of measurement points, and each pair of measurement points corresponding to the angle deviation are located on both sides of a notch of the wafer, the positioning angle deviation is calculated based on the differences in etched widths between each pair of measurement points in the corresponding one or more pairs of measurement points.
10. The wafer positioning system of claim 1, wherein the first channel is formed at an edge area of the second supporting area, and one or more chemical fluids flow between the first space and an exterior of the semiconductor processing device through a first through hole located in the upper chamber; and/or the first channel is formed at an edge area of the first supporting area, and the lower chamber comprises a first through hole, through which one or more chemical fluids flow between the first space and the exterior of the semiconductor processing device.
11. A wafer positioning method, comprising: placing, by a robot, a wafer on a first supporting area of a lower chamber of a semiconductor processing device according to set or updated current position configuration parameters; etching an edge area of the wafer using the semiconductor processing device; performing image acquisition and recognition to the etched wafer placed based on the current position configuration parameters to obtain etched edge information of the wafer; calculating position deviation parameters of the current position configuration parameters using a positioning calibration module with the obtained etched edge information of the wafer; and

updating, by a control unit, the current position configuration parameters based on the position deviation parameters to obtain the updated current position configuration parameters, and controlling the robot to place the obtained wafer on the first supporting area of the lower chamber at a position corresponding to the updated current position configuration parameters.

12. The wafer positioning method of claim 11, wherein after the calculating position deviation parameters of the current position configuration parameters using a positioning calibration module, the method further comprises: determining whether the position deviation parameters meet a predetermined position deviation range; stopping updating the current position configuration parameters If so; and updating the current position configuration parameters based on the position deviation parameter If not, and recalculating the position deviation parameters after obtaining the etched edge of the wafer again with the updated current position configuration parameters until the position deviation parameters corresponding to the latest current position parameters meet the predetermined position deviation range.

13. The wafer positioning method of claim 11, wherein the positioning calibration module calculates etched widths at multiple measurement points of the wafer using the etched edge information obtained through image recognition, and calculates the position deviation parameters based on the etched widths at the multiple measurement points.

14. The wafer positioning method of claim 13, wherein the position deviation parameter comprises a first direction distance deviation and a second direction distance deviation, while the current position configuration parameter comprises a first direction position coordinate and a second direction position coordinate, with the first direction and the second direction perpendicular to each other; the updated first direction position coordinate in the current position configuration parameter is obtained base on the first direction position coordinate in the current position configuration parameter and the first direction distance deviation in the position deviation parameter, the updated second direction position coordinate of the current position configuration parameter is obtained base on the second direction position coordinate in the current position configuration parameter and the second direction distance deviation in the position deviation parameter.

15. The wafer positioning method of claim 14, wherein the multiple measurement points comprise multiple pairs of measurement points, the first direction distance deviation of the position deviation parameter corresponds to one or more pairs of measurement points, a straight line connecting each pair of measurement points corresponding to the first direction distance deviation passes through the center of the wafer, and the first direction distance deviation is calculated based on the differences in the etched widths between each pair of measurement points in the corresponding one or more pairs of measurement points, alternatively, the second direction distance deviation of the position deviation parameters corresponds to one or more pairs of measurement points, and a straight line connecting each pair of measurement points corresponding to the second direction distance deviation passes through the center of the wafer, and the second direction distance deviation is calculated based on the differences in the etched widths between each pair of measurement points in the corresponding one or more pairs of measurement points.

16. The wafer positioning method of claim 14, wherein the position deviation parameter comprises a positioning angle deviation, the current position configuration parameter comprises a positioning angle, an updated positioning angle in the current configuration parameter is obtained based on the positioning angle in the current position configuration parameter and the positioning angle deviation in the position deviation parameter, and the positioning angle deviation of the position deviation parameter refers to a positioning angle deviation between the wafer and the first supporting area, the positioning angle deviation of the position deviation parameters corresponds to multiple measurement points, and a fitted straight-line slope of the multiple measurement points is calculated based on the differences in the etched widths and distances between the corresponding multiple measurement points, the positioning angle deviation is derived based on the fitted straight-line slope; or the positioning angle deviation of the position deviation parameters corresponds to

one or more pairs of measurement points, and each pair of measurement points corresponding to the angle deviation are located on both sides of a notch of the wafer, the positioning angle deviation is calculated based on the differences in the etched widths between each pair of measurement points in the corresponding one or more pairs of measurement points.

17. The wafer positioning method of claim 11, wherein a semiconductor processing device comprises a lower chamber with a first supporting area and an upper chamber with a second supporting area, wherein in response to the upper chamber engages with the lower chamber with the wafer placed between the first supporting area and the second supporting area, a first channel is formed at an edge area of the first supporting area and/or the second supporting area, wherein the first channel is configured to provide a first space for flow of one or more chemical fluids for etching the edge area of the wafer.
